Final Experiment Report

Synthetic Satellite Segmentation Project - LoveDA Rural

Goal
Build a reproducible semantic-segmentation pipeline and test whether controlled data interventions improve a compact U-Net trained on rural satellite imagery.

Dataset
LoveDA Rural: 1,366 training image/mask pairs and 992 validation pairs. Seven trainable land-cover classes; No Data pixels ignored.

What was built

Component	Completed work
Data pipeline	PNG image/mask loader, LoveDA label remapping, ignore label handling, and dataset statistics.
Training pipeline	Compact U-Net, CrossEntropyLoss(ignore_index=255), Adam, checkpoints, reproducible seeds, and saved configurations.
Evaluation	mIoU, per-class IoU, pixel accuracy, validation visualizations, controlled brightness robustness evaluation, and experiment reports.
Experiment protocol	One changed variable at a time; fixed 512 x 512 images, batch size 1, 15 epochs, learning rate 1e-4, and 1,366 updates per epoch.

Reference baseline

Real-only baseline: averaged 24.64% mIoU across seeds 42, 7, and 123. This frozen setup is the fair reference for every intervention.

Interventions Tested

Controlled experiments were judged by repeated seeds, not by one lucky training run.

Intervention	Evidence	Decision
10% brightness copies	Three seeds: clean mIoU fell 0.94 points on average. Brightness-shift gains were small and inconsistent by seed.	Rejected for this setup.
10% 90-degree rotations	Seed-42 pilot: mIoU fell from 24.42% to 21.54%.	Rejected early; no more seeds.
Barren-targeted oversampling	Three seeds: Barren IoU rose from 0.00% to 3.12% average while mIoU changed by only -0.04 points.	Retained as a rare-class tradeoff.

Why rare-class targeting was needed

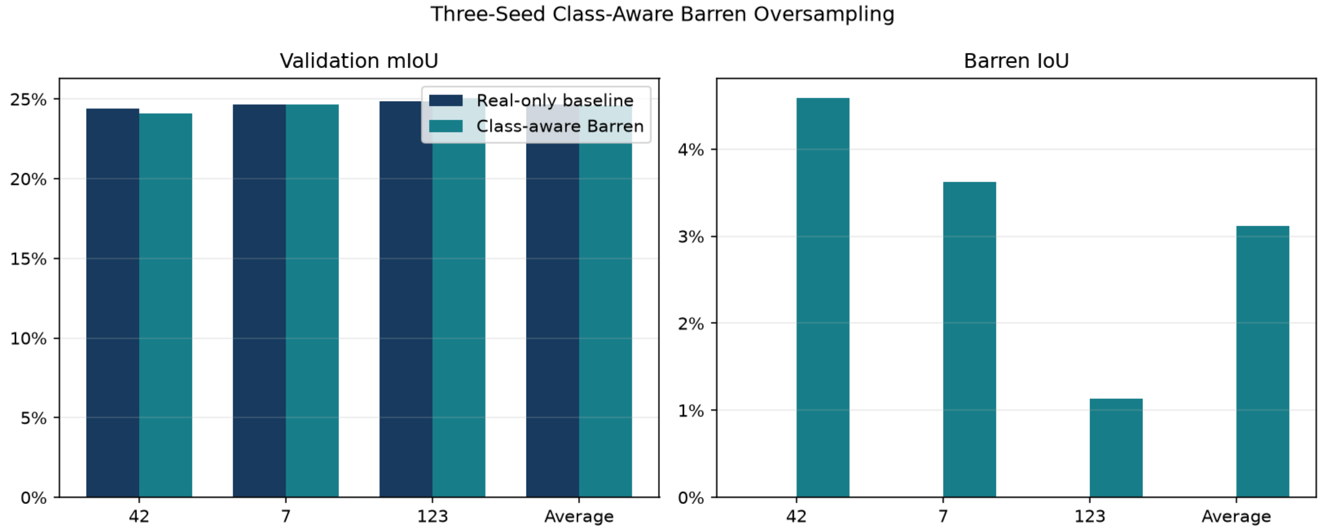
The Barren class covered only 3.17% of Rural training pixels. The real-only baseline produced 0.00% Barren IoU in every seed, meaning it did not successfully segment that class at all.

The class-aware sampler left masks and images unchanged. It simply sampled Barren-containing images with weight 4.0, increasing their expected share of training draws while preserving the 1,366-update budget.

This is not a claim that the model became better at every class. It is a controlled tradeoff aimed at a known failure case.

Final Barren Result

Three paired seeds: real-only baseline versus class-aware Barren oversampling.



Metric	Real-only average	Class-aware average	Change
Validation mIoU	24.64%	24.60%	-0.04%
Barren IoU	0.00%	3.12%	+3.12%
Pixel accuracy	53.83%	49.86%	-3.97%

Interpretation

Supported conclusion: Barren-targeted oversampling consistently makes the model predict the previously absent Barren class, with negligible average mIoU change. Important cost: pixel accuracy is 3.97 points lower, so this should be presented as a rare-class recall tradeoff rather than an unconditional performance improvement.

Project Conclusion

What this completed research phase establishes.

What we learned

- A complete segmentation research workflow was built and tested end to end.

- Generic image transformations are not automatically useful: both brightness and rotation failed to improve this compact model under the chosen conditions.

- Repeated seeds changed the interpretation of the brightness experiment, which is why replication was necessary.

- Class-aware oversampling addressed a specific failure: nonzero Barren IoU in all three seeds with almost unchanged mIoU.

Recommended status

This experiment phase is complete.
Keep the real-only model as the general reference baseline. Use class-aware Barren oversampling only when recovering the rare Barren class is worth the measured pixel-accuracy tradeoff. Do not continue tuning brightness or rotation in this project phase.

Possible future work

A future Phase 2 could test class-weighted loss or true copy-paste synthesis of Barren regions. Those would be separate research questions and should reuse the same fixed baseline and repeated-seed protocol.

Primary sources: saved experiment configurations, checkpoints, and metrics under models/; evaluation/class_aware_barren_comparison/results.json.